

Traffic Analysis Exercise (Wireshark)

Introduction – BSF-Nutrifeed

As part of the BSF-Nutrifeed Secure Infrastructure project, this traffic analysis exercise was conducted to examine how data flows within a simulated network environment supporting digital health and sustainable agricultural systems. The BSF-Nutrifeed ecosystem, aligned with SDG 3 (Good Health and Well-being), relies on digital platforms for monitoring, reporting, and system interactions; therefore, understanding network behavior is essential for ensuring confidentiality, integrity, and availability of its services.

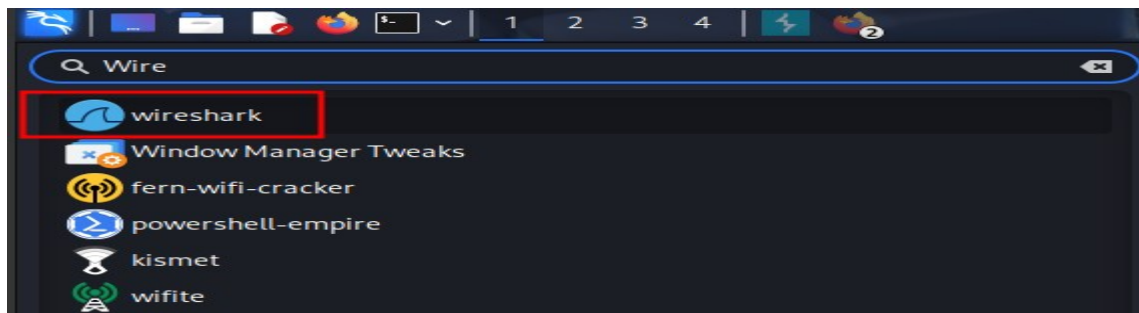
Using Wireshark, network traffic was captured and analyzed within a controlled virtual environment to observe communication between client and server systems. This exercise focused on identifying patterns of normal network behavior, such as HTTP requests and TCP session flows, while also assessing potential anomalies or unusual activities that could indicate misconfigurations or security weaknesses.

The analysis provides insight into how data is transmitted across the BSF-Nutrifeed infrastructure and highlights the importance of continuous monitoring in detecting suspicious behavior, preventing unauthorized access, and strengthening overall system resilience.

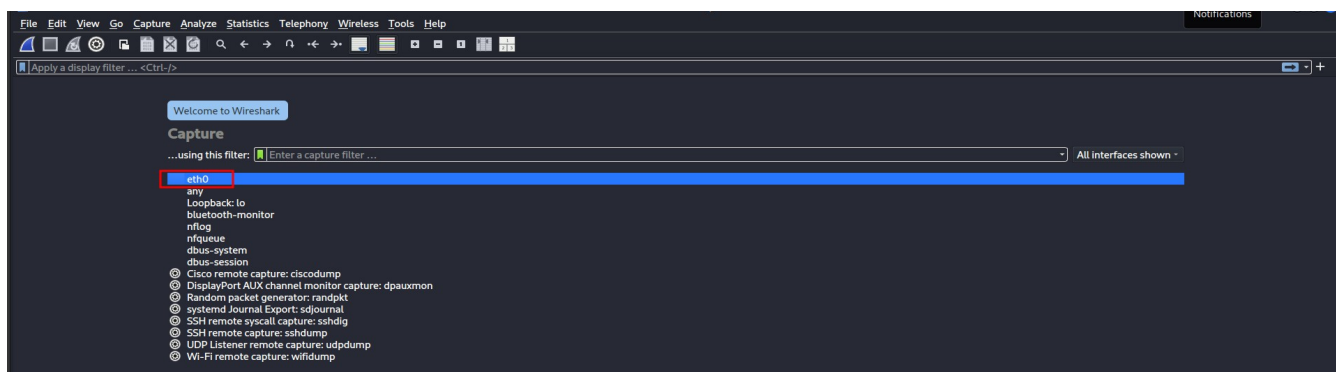
Hands-On

1. Wireshark

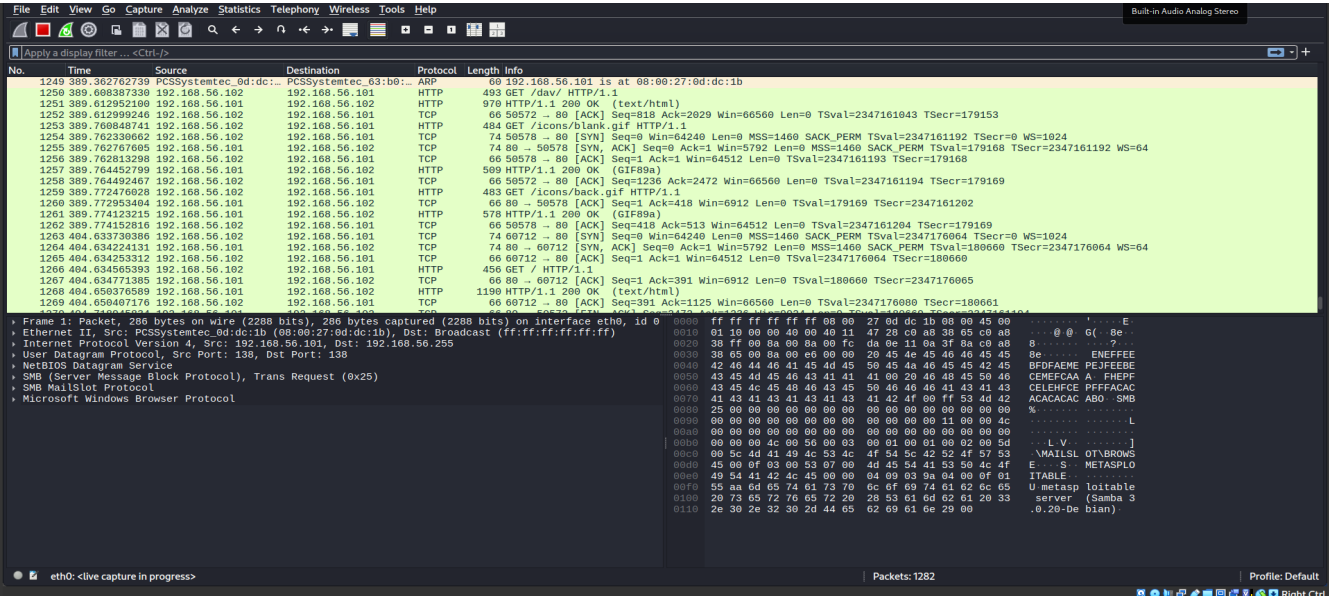
i) Open Wireshark



ii) Select the correct Network Interface

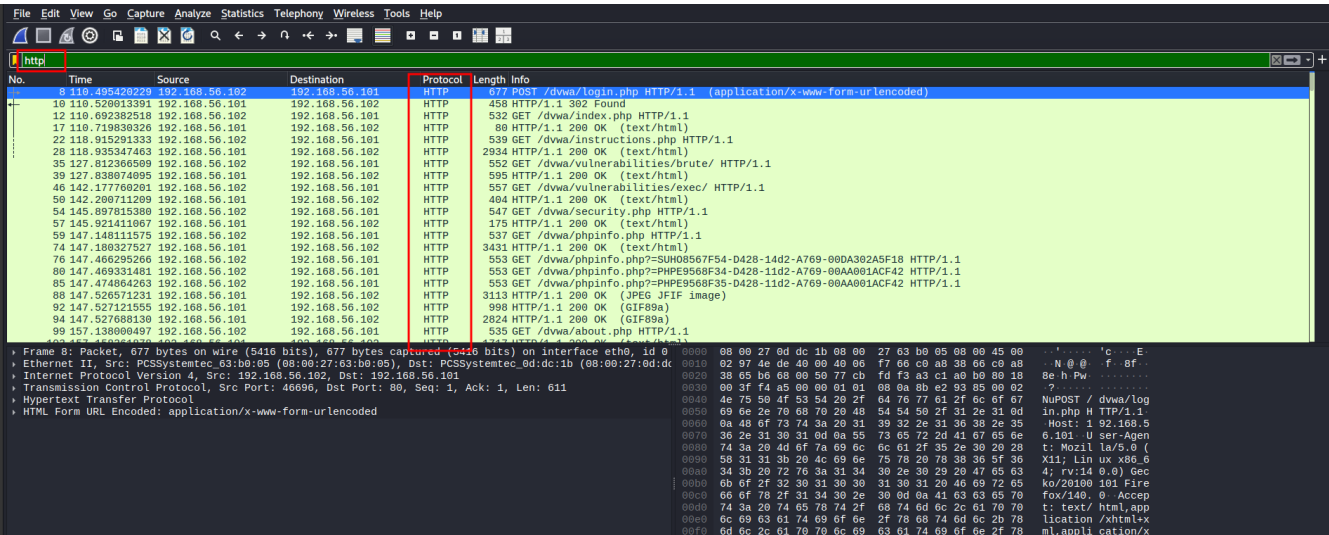


iii) Start Traffic Capturing Packets

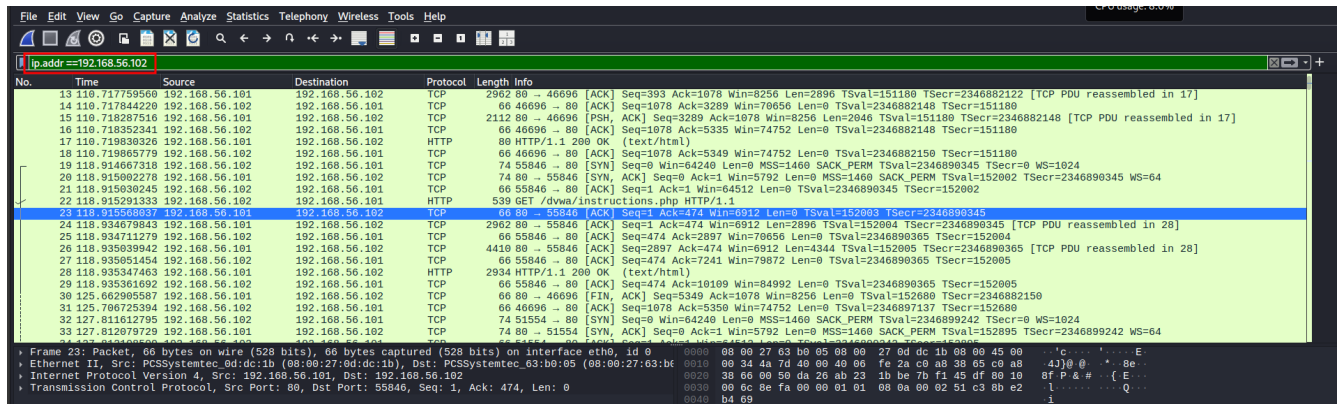


2. Traffic Analysis
Filtering & Inspection

i) HTTP Traffic:



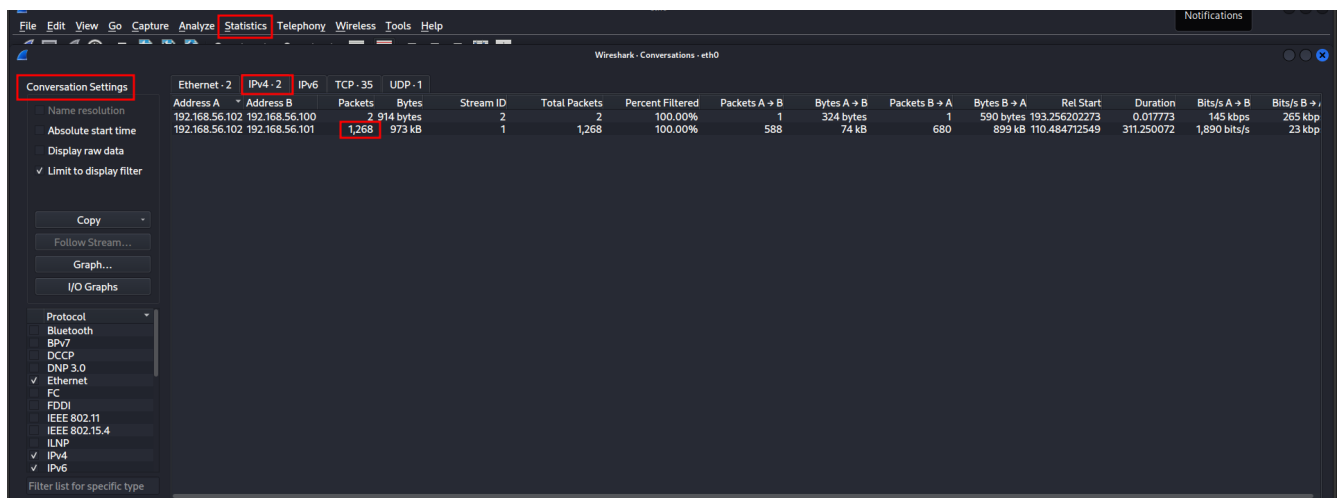
ii) My Kali IP Traffic (src & dst)



No.	Time	Source	Destination	Protocol	Length	Info
13	110.717759560	192.168.56.101	192.168.56.102	TCP	2562	88 → 46696 [ACK] Seq=393 Ack=1078 Win=8256 Len=2896 TSval=151180 TSecr=2346882122 [TCP PDU reassembled in 17]
14	110.717844220	192.168.56.102	192.168.56.101	TCP	66	46696 → 80 [ACK] Seq=1078 Ack=3289 Win=70656 Len=0 TSval=2346882148 TSecr=151180
15	110.718287516	192.168.56.101	192.168.56.102	TCP	2112	80 → 46696 [PSH, ACK] Seq=3289 Ack=1078 Win=8256 Len=2046 TSval=151180 TSecr=2346882148 [TCP PDU reassembled in 17]
16	110.718352341	192.168.56.102	192.168.56.101	TCP	66	46696 → 80 [ACK] Seq=1078 Ack=5335 Win=74752 Len=0 TSval=2346882148 TSecr=151180
17	110.719838326	192.168.56.101	192.168.56.102	HTTP	80	HTTP/1.1 200 OK [text/html]
18	110.719865779	192.168.56.102	192.168.56.101	TCP	66	46696 → 80 [ACK] Seq=1078 Ack=5349 Win=74752 Len=0 TSval=2346882150 TSecr=151180
19	118.914667318	192.168.56.102	192.168.56.101	TCP	74	55846 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2346890345 TSecr=0 WS=1024
20	118.915002278	192.168.56.101	192.168.56.102	TCP	74	80 → 55846 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=152002 TSecr=2346890345 WS=64
21	118.915083025	192.168.56.102	192.168.56.101	TCP	66	55846 → 80 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=2346890345 TSecr=152002
22	118.915291333	192.168.56.102	192.168.56.101	HTTP	539	GET /dms/instructions.php HTTP/1.1
23	110.915558637	192.168.56.101	192.168.56.102	TCP	66	80 → 55846 [ACK] Seq=1 Ack=474 Win=6912 Len=0 TSval=152004 TSecr=2346890345
24	118.934679043	192.168.56.101	192.168.56.102	TCP	2962	80 → 55846 [ACK] Seq=1 Ack=474 Win=6912 Len=2896 TSval=152004 TSecr=2346890345 [TCP PDU reassembled in 28]
25	118.934711279	192.168.56.102	192.168.56.101	TCP	66	55846 → 80 [ACK] Seq=474 Ack=2897 Win=70656 Len=0 TSval=2346890365 TSecr=152004
26	118.935039942	192.168.56.101	192.168.56.102	TCP	4410	80 → 55846 [ACK] Seq=2897 Ack=474 Win=6912 Len=4344 TSval=152005 TSecr=2346890365 [TCP PDU reassembled in 28]
27	118.935051454	192.168.56.102	192.168.56.101	TCP	66	55846 → 80 [ACK] Seq=474 Ack=7241 Win=79872 Len=0 TSval=2346890365 TSecr=152005
28	118.935347463	192.168.56.101	192.168.56.102	HTTP	2934	HTTP/1.1 200 OK [text/html]
29	118.935361692	192.168.56.102	192.168.56.101	TCP	66	55846 → 80 [ACK] Seq=474 Ack=10109 Win=84992 Len=0 TSval=2346890365 TSecr=152005
30	125.662995587	192.168.56.101	192.168.56.102	TCP	66	80 → 46696 [FIN, ACK] Seq=5349 Ack=1078 Win=8256 Len=0 TSval=152680 TSecr=2346882150
31	125.706725394	192.168.56.102	192.168.56.101	TCP	66	46696 → 80 [ACK] Seq=1078 Ack=5350 Win=74752 Len=0 TSval=2346897137 TSecr=152680
32	127.816127295	192.168.56.102	192.168.56.101	TCP	74	51554 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2346899242 TSecr=0 WS=1024
33	127.812679729	192.168.56.101	192.168.56.102	TCP	74	80 → 51554 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=152895 TSecr=2346899242 WS=64

iii) Repeated Connections

- Same IP appearing many times (Statistics → Conversation → IPv4)



Name resolution	Address A	Address B	Packets	Bytes	Stream ID	Total Packets	Percent Filtered	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
Absolute start time	192.168.56.102	192.168.56.100	2	914 bytes	2	2	100.00%	1	324 bytes	1	590 bytes	193.256202273	0.017773	145 kbps	265 kbp
Display raw data	192.168.56.102	192.168.56.101	1,268	973 kB	1	1,268	100.00%	588	74 kB	680	899 kB	110.484712549	311.250072	1,890 bits/s	23 kbp

Comment:

Analysis of network conversations revealed repeated communication between 192.168.56.102 and 192.168.56.101, with a high number of exchanged packets, indicating continuous interaction.

- Same Request Repeated

No.	Time	Source	Destination	Protocol	Length	Info
50	142.200711299	192.168.56.101	192.168.56.102	HTTP	404	HTTP/1.1 200 OK (text/html)
54	145.897815380	192.168.56.102	192.168.56.101	HTTP	547	GET /dvwa/security.php HTTP/1.1
57	145.921411067	192.168.56.101	192.168.56.102	HTTP	175	HTTP/1.1 200 OK (text/html)
59	147.148311575	192.168.56.102	192.168.56.101	HTTP	537	GET /dvwa/phpinfo.php HTTP/1.1
74	147.189327527	192.168.56.101	192.168.56.102	HTTP	3431	HTTP/1.1 200 OK (text/html)
76	147.466295266	192.168.56.102	192.168.56.101	HTTP	553	GET /dvwa/phpinfo.php?SUH08567F54-0428-14d2-A769-06DA302A5F18 HTTP/1.1
80	147.469331461	192.168.56.102	192.168.56.101	HTTP	553	GET /dvwa/phpinfo.php?PHPE9568F35-0428-11d2-A769-06AA081ACF42 HTTP/1.1
85	147.474864293	192.168.56.102	192.168.56.101	HTTP	553	GET /dvwa/phpinfo.php?PHPE9568F35-0428-11d2-A769-06AA081ACF42 HTTP/1.1
88	147.526511231	192.168.56.101	192.168.56.102	HTTP	3113	HTTP/1.1 200 OK (JPEG JFIF image)
92	147.527121555	192.168.56.101	192.168.56.102	HTTP	998	HTTP/1.1 200 OK (GIF89a)
94	147.527686139	192.168.56.101	192.168.56.102	HTTP	2824	HTTP/1.1 200 OK (GIF89a)
99	157.139809497	192.168.56.102	192.168.56.101	HTTP	535	GET /dvwa/about.php HTTP/1.1
103	157.158361878	192.168.56.101	192.168.56.102	HTTP	1717	HTTP/1.1 200 OK (text/html)
114	167.348783235	192.168.56.102	192.168.56.101	HTTP	559	GET /dvwa/vulnerabilities/fi/?page=include.php HTTP/1.1
118	167.373520663	192.168.56.101	192.168.56.102	HTTP	156	HTTP/1.1 200 OK (text/html)
128	188.941138358	192.168.56.102	192.168.56.101	HTTP	559	GET /dvwa/setup.php HTTP/1.1
130	188.959038290	192.168.56.101	192.168.56.102	HTTP	3968	HTTP/1.1 200 OK (text/html)
132	189.032243044	192.168.56.102	192.168.56.101	HTTP	522	GET /dvwa/images/spanner.png HTTP/1.1
133	189.034031930	192.168.56.101	192.168.56.102	HTTP	1078	HTTP/1.1 200 OK (PNG)
142	195.158850561	192.168.56.102	192.168.56.101	HTTP	534	GET /dvwa/phpinfo.php HTTP/1.1
156	195.185193181	192.168.56.101	192.168.56.102	HTTP	3423	HTTP/1.1 200 OK (text/html)
* Frame 88: Packet, 3113 bytes on wire (24984 bits), 3113 bytes captured (24984 bits) on interface eth0, 0000 08 00 27 63 b0 05 08 00 27 0d dc 1b 08 00 45 00 ...c... ..E. * Ethernet II, Src: PCSSystemtec_0d:dc:1b (08:00:27:0d:dc:1b), Dst: PCSSystemtec_63:b0:05 (08:00:27:63:b0:05) ...@ @ B> 8e... * Internet Protocol Version 4, Src: 192.168.56.101, Dst: 192.168.56.102 8f P...u 1ec... * Transmission Control Protocol, Src Port: 80, Dst Port: 35476, Seq: 60285, Ack: 1931, Len: 3847V... * Hypertext Transfer Protocol # HTTP/1.1 200 O * JPEG File Interchange Format K. Date: Mon, 30 0000 08 00 27 63 b0 05 08 00 27 0d dc 1b 08 00 45 00 ...c... ..E. 0010 0c 1b fa 82 40 09 40 06 42 3e c0 a8 38 65 c0 a8 ...@ @ B> 8e... 0020 38 66 00 50 8a c1 75 8c 31 65 43 a2 a1 80 18 8f P...u 1ec... 0030 00 9e fe 29 00 00 01 01 08 0a 00 02 5c ff 8b e3V... 0040 23 f9 48 54 54 50 2f 31 2e 31 20 32 30 30 20 4f # HTTP/1.1 200 O 0050 4b 0d 0a 44 61 74 65 3a 20 4d 6f 6e 2c 20 33 30 K. Date: Mon, 30 0060 20 4d 61 72 20 32 30 32 30 20 31 32 3a 35 35 3a Mar 202 0 12:55: 0070 30 30 20 47 4d 54 0d 0a 83 65 72 76 65 72 3a 20 00 GMT Server: 0080 41 70 61 63 68 65 2f 32 2e 32 2e 38 20 28 55 62 Apache/2.2.8 (Ub 0090 75 6e 74 75 29 20 44 41 56 2f 32 0d 0a 58 2d 50 untu) DA V/2. X-P 00a0 6f 77 65 72 65 64 2d 42 79 3a 20 50 48 50 2f 35 overed-B y: PHP/5 00b0 2e 32 2e 34 2d 32 75 62 75 6e 74 75 35 2e 31 30 2.4-2ub untu5.10 00c0 0d 0a 43 6f 6e 74 65 6e 74 2d 4c 65 6e 67 74 68 Content-TLength 00d0 3a 20 32 38 31 33 0d 0a 4b 65 65 70 2d 41 6c 69 : 2813 Keep-All 00e0 6f 77 65 72 65 64 2d 42 79 3a 20 50 48 50 2f 35 ves time out=15 00f0 6d 61 78 30 39 37 0d 0a 43 6f 6e 6e 65 63 74 69 max=97 Connecti 0100 6f 6e 3a 29 4b 65 65 70 2d 41 6c 69 76 65 0d 0a on: Keep-Alive 0110 43 6f 6e 74 65 6e 74 2d 54 79 70 65 3a 20 69 6d Content-Type: im 0120 61 67 65 2f 6a 70 65 67 0d 0a 0a ff 6b ff 6b age/jpeg 0130 00 10 4a 46 49 46 00 01 01 01 00 48 00 48 00 00 JFIF ...H... 0140 ff e1 00 16 45 78 69 66 00 0d 4d 4d 00 2a 00 00 ...Exif ...MM*... 0150 00 08 00 00 00 00 00 00 ff db 00 43 00 01 01 01C...						
No.	Time	Source	Destination	Protocol	Length	Info
319	248.952758399	192.168.56.102	192.168.56.101	HTTP	1629	POST /phpMyAdmin/index.php HTTP/1.1 (application/x-www-form-urlencoded)
321	249.069980628	192.168.56.101	192.168.56.102	HTTP	1646	HTTP/1.1 302 Found
323	249.117822639	192.168.56.102	192.168.56.101	HTTP	706	GET /phpMyAdmin/index.php?token=abe0aeaa440ac67d659421d177684336 HTTP/1.1
406	249.273956910	192.168.56.101	192.168.56.102	HTTP	71	HTTP/1.1 200 OK (text/html)
411	249.411176215	192.168.56.102	192.168.56.101	HTTP	729	GET /phpMyAdmin/phpmyadmin.css.php?token=abe0aeaa440ac67d659421d177684336&js_frame=right&nocache=2457687151 HTTP/1.1
553	249.484639155	192.168.56.101	192.168.56.102	HTTP	71	HTTP/1.1 200 OK (text/css)
555	249.518006085	192.168.56.102	192.168.56.101	HTTP	746	GET /phpMyAdmin/themes/original/img/s_error.png HTTP/1.1
558	249.536113857	192.168.56.101	192.168.56.102	HTTP	634	HTTP/1.1 200 OK (PNG)
568	264.936234112	192.168.56.102	192.168.56.101	HTTP	456	GET / HTTP/1.1
578	264.95553206	192.168.56.101	192.168.56.102	HTTP	1198	HTTP/1.1 200 OK (text/html)
572	267.386290782	192.168.56.102	192.168.56.101	HTTP	500	GET /mutillidae/ HTTP/1.1
658	267.499161973	192.168.56.101	192.168.56.102	HTTP	71	HTTP/1.1 200 OK (text/html)
661	267.681397222	192.168.56.102	192.168.56.101	HTTP	457	GET /mutillidae/styles/global-styles.css HTTP/1.1
665	267.684080627	192.168.56.102	192.168.56.101	HTTP	469	GET /mutillidae/styles/ddsmoothmenu/ddsmoothmenu.css HTTP/1.1
660	267.684231272	192.168.56.101	192.168.56.102	HTTP	8097	HTTP/1.1 200 OK (text/css)
669	267.685356718	192.168.56.102	192.168.56.101	HTTP	445	GET /mutillidae/javascript/bookmark-site.js HTTP/1.1
670	267.685451449	192.168.56.101	192.168.56.102	HTTP	2613	HTTP/1.1 200 OK (text/css)
672	267.686558747	192.168.56.101	192.168.56.102	HTTP	1442	HTTP/1.1 200 OK (application/x-javascript)
676	267.698661733	192.168.56.102	192.168.56.101	HTTP	457	GET /mutillidae/javascript/ddsmoothmenu/ddsmoothmenu.js HTTP/1.1
680	267.698293617	192.168.56.101	192.168.56.102	HTTP	4674	HTTP/1.1 200 OK (application/x-javascript)
685	267.709123666	192.168.56.102	192.168.56.101	HTTP	471	GET /mutillidae/styles/ddsmoothmenu/ddsmoothmenu.v.css HTTP/1.1
* Frame 88: Packet, 3113 bytes on wire (24984 bits), 3113 bytes captured (24984 bits) on interface eth0, 0000 08 00 27 63 b0 05 08 00 27 0d dc 1b 08 00 45 00 ...c... ..E. * Ethernet II, Src: PCSSystemtec_0d:dc:1b (08:00:27:0d:dc:1b), Dst: PCSSystemtec_63:b0:05 (08:00:27:63:b0:05) ...@ @ B> 8e... * Internet Protocol Version 4, Src: 192.168.56.101, Dst: 192.168.56.102 8f P...u 1ec... * Transmission Control Protocol, Src Port: 80, Dst Port: 35476, Seq: 60285, Ack: 1931, Len: 3847V... * Hypertext Transfer Protocol # HTTP/1.1 200 O * JPEG File Interchange Format K. Date: Mon, 30 0000 08 00 27 63 b0 05 08 00 27 0d dc 1b 08 00 45 00 ...c... ..E. 0010 0c 1b fa 82 40 09 40 06 42 3e c0 a8 38 65 c0 a8 ...@ @ B> 8e... 0020 38 66 00 50 8a c1 75 8c 31 65 43 a2 a1 80 18 8f P...u 1ec... 0030 00 9e fe 29 00 00 01 01 08 0a 00 02 5c ff 8b e3V... 0040 23 f9 48 54 54 50 2f 31 2e 31 20 32 30 30 20 4f # HTTP/1.1 200 O 0050 4b 0d 0a 44 61 74 65 3a 20 4d 6f 6e 2c 20 33 30 K. Date: Mon, 30 0060 20 4d 61 72 20 32 30 32 30 20 31 32 3a 35 35 3a Mar 202 0 12:55: 0070 30 30 20 47 4d 54 0d 0a 83 65 72 76 65 72 3a 20 00 GMT Server: 0080 41 70 61 63 68 65 2f 32 2e 32 2e 38 20 28 55 62 Apache/2.2.8 (Ub 0090 75 6e 74 75 29 20 44 41 56 2f 32 0d 0a 58 2d 50 untu) DA V/2. X-P 00a0 6f 77 65 72 65 64 2d 42 79 3a 20 50 48 50 2f 35 overed-B y: PHP/5 00b0 2e 32 2e 34 2d 32 75 62 75 6e 74 75 35 2e 31 30 2.4-2ub untu5.10						

Comment:

Multiple identical HTTP GET requests to /dvwa/phpinfo.php?, /phpMyAdmin/index.php and /HTTP/1.1, were observed, indicating repeated page access or refresh activity.

No communication with external IP was captured.

iv) Traffic Pattern

- Many GET/POST requests

No.	Time	Source	Destination	Protocol	Length	Info
1	110.495429229	192.168.56.102	192.168.56.101	HTTP	677	POST /dwa/login.php HTTP/1.1 (application/x-www-form-urlencoded)
10	110.520013391	192.168.56.101	192.168.56.102	HTTP	458	HTTP/1.1 302 Found
12	110.692382518	192.168.56.102	192.168.56.101	HTTP	532	GET /dwa/index.php HTTP/1.1
17	110.719830326	192.168.56.101	192.168.56.102	HTTP	80	HTTP/1.1 200 OK (text/html)
22	110.915291333	192.168.56.102	192.168.56.101	HTTP	539	GET /dwa/instructions.php HTTP/1.1
28	110.935347463	192.168.56.101	192.168.56.102	HTTP	2934	HTTP/1.1 200 OK (text/html)
35	127.812366509	192.168.56.102	192.168.56.101	HTTP	552	GET /dwa/vulnerabilities/brute/ HTTP/1.1
39	127.839074095	192.168.56.101	192.168.56.102	HTTP	595	HTTP/1.1 200 OK (text/html)
40	142.177662091	192.168.56.102	192.168.56.101	HTTP	557	GET /dwa/vulnerabilities/exec/ HTTP/1.1
50	142.200711209	192.168.56.101	192.168.56.102	HTTP	404	HTTP/1.1 200 OK (text/html)
54	145.897815300	192.168.56.102	192.168.56.101	HTTP	547	GET /dwa/security.php HTTP/1.1
57	145.921411067	192.168.56.101	192.168.56.102	HTTP	175	HTTP/1.1 200 OK (text/html)
59	147.148111575	192.168.56.102	192.168.56.101	HTTP	537	GET /dwa/phpinfo.php HTTP/1.1
74	147.180327527	192.168.56.101	192.168.56.102	HTTP	3431	HTTP/1.1 200 OK (text/html)
76	147.460295266	192.168.56.102	192.168.56.101	HTTP	553	GET /dwa/phpinfo.php?SUH08567F54-D428-14d2-A769-00DA302A5F18 HTTP/1.1
80	147.469331481	192.168.56.102	192.168.56.101	HTTP	553	GET /dwa/phpinfo.php?PHP9568F34-D428-11d2-A769-00AA081ACF42 HTTP/1.1
85	147.474864263	192.168.56.102	192.168.56.101	HTTP	553	GET /dwa/phpinfo.php?PHP9568F34-D428-11d2-A769-00AA081ACF42 HTTP/1.1
88	147.526571231	192.168.56.101	192.168.56.102	HTTP	3113	HTTP/1.1 200 OK (JPEG JFIF image)
92	147.527121555	192.168.56.101	192.168.56.102	HTTP	998	HTTP/1.1 200 OK (GIF89a)
94	147.527689130	192.168.56.101	192.168.56.102	HTTP	2824	HTTP/1.1 200 OK (GIF89a)
99	157.138060497	192.168.56.102	192.168.56.101	HTTP	466	GET /dwa/about.php HTTP/1.1

Comment:

Traffic analysis revealed multiple HTTP GET requests between the client and the web server during browsing activity. Only one POST request was observed within the capture period, indicating that one form submission or data upload occurred during the session.

- TCP handshake patterns

No.	Time	Source	Destination	Protocol	Length	Info
68	147.179690995	192.168.56.101	192.168.56.102	TCP	10202	80 → 35476 [ACK] Seq=35290 Ack=1444 Win=9024 Len=0 TSval=154837 TSecr=2346918609 [TCP PDU reassembled in 74]
69	147.179710580	192.168.56.102	192.168.56.101	TCP	66	35476 → 80 [ACK] Seq=1444 Ack=4536 Win=66416 Len=0 TSval=2346918610 TSecr=154837
70	147.180064595	192.168.56.101	192.168.56.102	TCP	10202	80 → 35476 [PSH, ACK] Seq=45336 Ack=1444 Win=9024 Len=10136 TSval=154837 TSecr=2346918610 [TCP PDU reassembled in 74]
71	147.180086928	192.168.56.102	192.168.56.101	TCP	66	35476 → 80 [ACK] Seq=1444 Ack=55472 Win=84992 Len=0 TSval=2346918610 TSecr=154837
72	147.180214325	192.168.56.101	192.168.56.102	TCP	1514	80 → 35476 [ACK] Seq=55472 Ack=1444 Win=9024 Len=1448 TSval=154837 TSecr=2346918610 [TCP PDU reassembled in 74]
73	147.180230985	192.168.56.102	192.168.56.101	TCP	66	35476 → 80 [ACK] Seq=1444 Ack=56929 Win=93184 Len=0 TSval=2346918610 TSecr=154837
74	147.180327527	192.168.56.101	192.168.56.102	HTTP	3431	HTTP/1.1 200 OK (text/html)
75	147.180340691	192.168.56.102	192.168.56.101	TCP	66	35476 → 80 [ACK] Seq=1444 Ack=60285 Win=90112 Len=0 TSval=2346918610 TSecr=154837
76	147.460295266	192.168.56.102	192.168.56.101	HTTP	553	GET /dwa/phpinfo.php?SUH08567F54-D428-14d2-A769-00DA302A5F18 HTTP/1.1
77	147.460690305	192.168.56.102	192.168.56.101	TCP	74	35492 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2346918897 TSecr=0 WS=1024
78	147.460647278	192.168.56.101	192.168.56.102	TCP	74	80 → 35492 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=154865 TSecr=2346918897 WS=64
79	147.460678852	192.168.56.102	192.168.56.101	TCP	66	35492 → 80 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=2346918897 TSecr=154865
80	147.469331481	192.168.56.102	192.168.56.101	HTTP	553	GET /dwa/phpinfo.php?PHP9568F34-D428-11d2-A769-00AA081ACF42 HTTP/1.1
81	147.469852121	192.168.56.102	192.168.56.101	TCP	66	80 → 35492 [ACK] Seq=1 Ack=408 Win=6912 Len=0 TSval=154866 TSecr=2346918999
82	147.473971371	192.168.56.102	192.168.56.101	TCP	74	35506 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2346918904 TSecr=0 WS=1024
83	147.474331769	192.168.56.101	192.168.56.102	TCP	74	80 → 35506 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=154866 TSecr=2346918904 WS=64
84	147.474367000	192.168.56.102	192.168.56.101	TCP	66	35506 → 80 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=2346918904 TSecr=154866
85	147.474864263	192.168.56.102	192.168.56.101	HTTP	553	GET /dwa/phpinfo.php?PHP9568F34-D428-11d2-A769-00AA081ACF42 HTTP/1.1
86	147.475167619	192.168.56.102	192.168.56.101	TCP	66	80 → 35506 [ACK] Seq=1 Ack=488 Win=6912 Len=0 TSval=154866 TSecr=2346918905
87	147.501967361	192.168.56.101	192.168.56.102	TCP	66	80 → 35476 [ACK] Seq=60285 Ack=1931 Win=10112 Len=0 TSval=154869 TSecr=2346918996
88	147.526571231	192.168.56.101	192.168.56.102	HTTP	3113	HTTP/1.1 200 OK (JPEG JFIF image)

Comment:

TCP traffic analysis revealed SYN packets indicating connection initiation, ACK packets confirming communication, and PSH packets representing data transfer. This demonstrates a complete TCP session lifecycle between the client and server during web browsing activity.

- Continuous packet flow

No.	Time	Source	Destination	Protocol	Length	Info
306	234.553031097	192.168.56.102	192.168.56.101	HTTP	771	GET /phpMyAdmin/themes/original/img/s_warn.png HTTP/1.1
307	234.554056089	192.168.56.101	192.168.56.102	HTTP	623	HTTP/1.1 200 OK (PNG)
308	234.590626038	192.168.56.101	192.168.56.101	HTTP	626	GET /phpMyAdmin/favicon.ico HTTP/1.1
309	234.592404757	192.168.56.101	192.168.56.102	TCP	7396	88 → 47458 [ACK] Seq=38175 Ack=3764 Win=14400 Len=7240 TSval=163602 TSecr=2347086021 [TCP PDU reassembled in 312]
310	234.592523598	192.168.56.102	192.168.56.101	TCP	66	47458 → 88 [ACK] Seq=3764 Ack=43415 Win=88064 Len=0 TSval=2347086022 TSecr=163602
311	234.592735282	192.168.56.101	192.168.56.102	TCP	7396	88 → 47458 [ACK] Seq=43415 Ack=3764 Win=14400 Len=7240 TSval=163602 TSecr=2347086021 [TCP PDU reassembled in 312]
312	234.592997813	192.168.56.101	192.168.56.102	HTTP	4798	HTTP/1.1 200 OK (Image/x-icon)
313	234.593092870	192.168.56.101	192.168.56.101	TCP	66	47458 → 88 [ACK] Seq=3764 Ack=55379 Win=76800 Len=0 TSval=2347086023 TSecr=163602
314	235.604144066	192.168.56.101	192.168.56.102	TCP	66	88 → 47448 [FIN, ACK] Seq=2075 Ack=934 Win=7936 Len=0 TSval=163704 TSecr=2346992100
315	235.648663617	192.168.56.102	192.168.56.101	TCP	66	47448 → 88 [ACK] Seq=934 Ack=2076 Win=63488 Len=0 TSval=2347087079 TSecr=163704
316	249.952083977	192.168.56.102	192.168.56.101	TCP	74	46672 → 88 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2347020302 TSecr=0 WS=1024
317	249.952506996	192.168.56.101	192.168.56.102	TCP	74	88 → 46672 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=165043 TSecr=2347020382 WS=64
318	249.952535541	192.168.56.102	192.168.56.101	TCP	66	46672 → 88 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=2347020383 TSecr=165043
319	249.952758399	192.168.56.102	192.168.56.101	HTTP	1029	POST /phpMyAdmin/index.php HTTP/1.1 (application/x-www-form-urlencoded)
320	249.953019059	192.168.56.101	192.168.56.102	TCP	66	88 → 46672 [ACK] Seq=1 Ack=964 Win=7744 Len=0 TSval=165043 TSecr=2347020383
321	249.069986828	192.168.56.101	192.168.56.102	HTTP	1046	HTTP/1.1 302 Found
322	249.070046847	192.168.56.102	192.168.56.101	TCP	66	46672 → 88 [ACK] Seq=964 Ack=981 Win=64512 Len=0 TSval=2347020500 TSecr=165055
323	249.117822839	192.168.56.102	192.168.56.101	HTTP	706	GET /phpMyAdmin/index.php?token=abe0a4a40ac67d059421d177684336 HTTP/1.1
324	249.118290639	192.168.56.101	192.168.56.102	TCP	66	88 → 46672 [ACK] Seq=981 Ack=1604 Win=9664 Len=0 TSval=165060 TSecr=2347020548
325	249.262063890	192.168.56.101	192.168.56.102	TCP	1150	88 → 46672 [PSH, ACK] Seq=981 Ack=1604 Win=9664 Len=1084 TSval=165074 TSecr=2347020548 [TCP PDU reassembled in 406]
326	249.262014671	192.168.56.101	192.168.56.102	TCP	73	88 → 46672 [PSH, ACK] Seq=2005 Ack=1604 Win=9664 Len=7 TSval=165074 TSecr=2347020548 [TCP PDU reassembled in 406]

- Frame 319: Packet, 1029 bytes on wire (8232 bits), 1029 bytes captured (8232 bits) on interface eth0, 3
- Ethernet II, Src: PCSSystemtec_63:b0:05 (08:00:27:63:b0:05), Dst: PCSSystemtec_Bd:dc:1b (08:00:27:0d:dc:1b)
- Internet Protocol Version 4, Src: 192.168.56.102, Dst: 192.168.56.101
- Transmission Control Protocol, Src Port: 46672, Dst Port: 88, Seq: 1, Ack: 1, Len: 963
- Hypertext Transfer Protocol
- HTML Form URL Encoded: application/x-www-form-urlencoded

Comment:

A continuous flow of packets was observed throughout the capture period, indicating sustained network activity during browsing operations.

Traffic Analysis Summary

Network traffic was captured using Wireshark on the active interface (eth0) over a period of approximately 3 minutes. During the capture, web browsing activity was performed to generate HTTP traffic.

Analysis revealed multiple HTTP GET and a POST requests between the client and web servers, indicating normal browsing behavior. Repeated connections to the same IP addresses were observed, showing continuous communication between the client and target servers.

No external IP addresses were identified in the captured traffic. Also, No encrypted HTTPS traffic was analyzed in detail, highlighting potential exposure of sensitive data over HTTP.

Overall, the traffic pattern demonstrated typical client-server interaction, with identifiable request-response sequences.